CSE422 Lab Project Report

Customer Category Classification (Segmentation)

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1. Introduction

This project builds a **customer segmentation classifier** using the provided dataset `Customer_Category_Classifier_Datas`
Each customer belongs to one of **four segments** (A, B, C, D) stored in the target column **Segmentation**.

Aim: Predict the correct segment for a new customer from demographic/behavioural features.

Problem type: Multi-class classification (output is a category, not a continuous value).

2. Dataset Description

Dataset size: 8068 rows × 11 columns.

Columns:

- ID
- Gender
- Ever_Married
- Age
- Graduated
- Profession
- Work_Experience
- Spending_Score
- Family_Size
- Var_1
- Segmentation

2.1 Class Distribution (Imbalance Check)

Because the dataset is slightly imbalanced, we report **macro** metrics (macro precision/recall/F1) and use **stratified** splitting/CV.

Class	Count	Percent (%)
D	2268	28.11
A	1972	24.44
C	1970	24.42
B	1858	23.03

2.2 Missing Values

Missing values before preprocessing:

Column	Missing Count
Work_Experience	829
Family_Size	335
Ever_Married	140
Profession	124
Graduated	78
Var_1	76

Missing values after preprocessing: **0 remaining** (all handled).

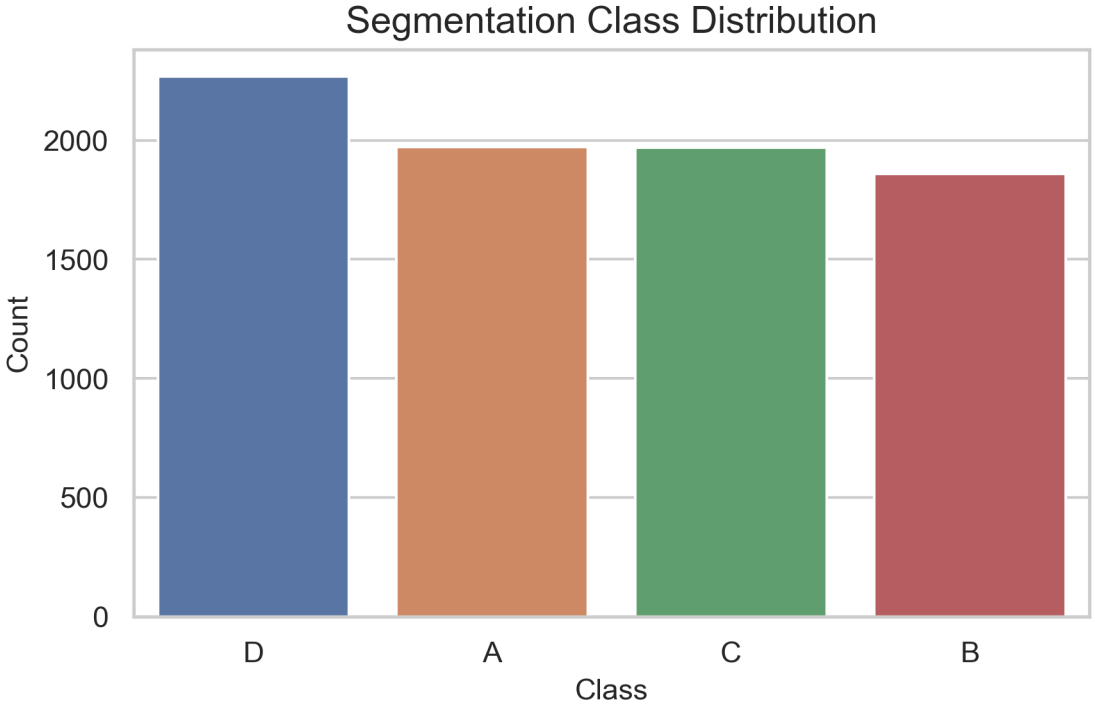


Figure 1: Segmentation Class Distribution

2.3 Exploratory Data Analysis (EDA)

Numeric feature summary by segment (count/mean/median):

('Segmentation', '')	('Age', 'count')	('Age', 'mean')	('Age', 'median')	('Work_Experience', 'count')	('Work_Experience', 'mean')	('Work_Experience', 'median')	('Family_Size', 'count')	('Family_Size', 'mean')	('Family_Size', 'median')
A	1972	44.92	41	1778	2.87	1	1877	2.44	2
B	1858	48.2	46	1666	2.38	1	1815	2.7	2
C	1970	49.14	49	1815	2.24	1	1926	2.97	3
D	2268	33.39	29	1980	3.02	1	2115	3.23	3

P(Ever_Married | Segmentation) (row-normalized):

Segmentation	No	Yes
A	0.413	0.587
B	0.262	0.738
C	0.204	0.796
D	0.725	0.275

P(Graduated | Segmentation) (row-normalized):

Segmentation	No	Yes
A	0.374	0.626
B	0.276	0.724
C	0.178	0.822
D	0.64	0.36

P(Spending_Score | Segmentation) (row-normalized):

Segmentation	Average	High	Low
A	0.174	0.137	0.689
B	0.318	0.207	0.476
C	0.458	0.206	0.336
D	0.061	0.069	0.87

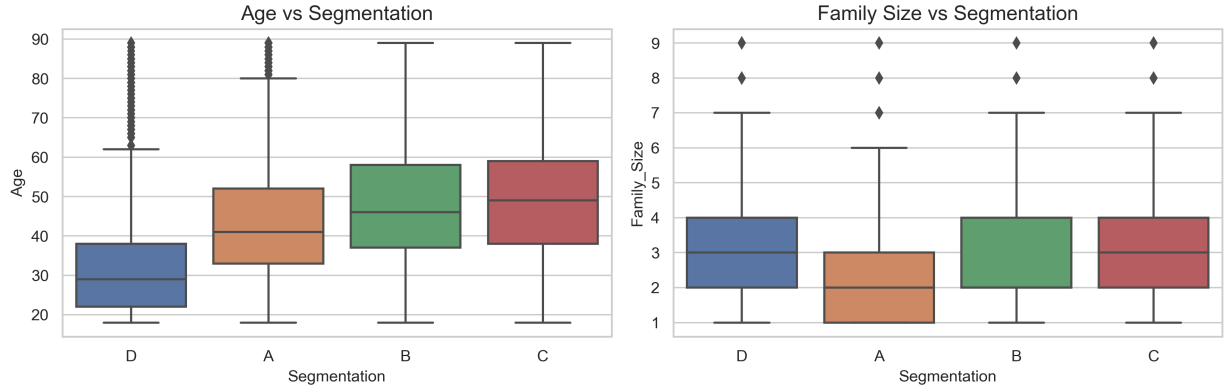


Figure 2: EDA Boxplots

2.4 Numeric Feature Correlation

Correlation is meaningful mainly for numeric columns; we plot a heatmap for `Age`, `Work_Experience`, and `Family_Size`.

3. Dataset Pre-processing

We prepare the data using a leakage-safe preprocessing pipeline:

- Drop `ID` because it is an identifier, not informative.
- Fill missing numeric values with the **median**.
- Fill missing categorical values with the **mode** (most frequent).
- One-hot encode categorical features.
- Standardize numeric features (important for KNN/SVM/Neural Net).

4. Dataset Splitting (Train/Test)

We split the dataset into **80% training** and **20% testing**, using **stratified** splitting so the class proportions stay similar.

- Training set size: (6454, 9)
- Test set size: (1614, 9)

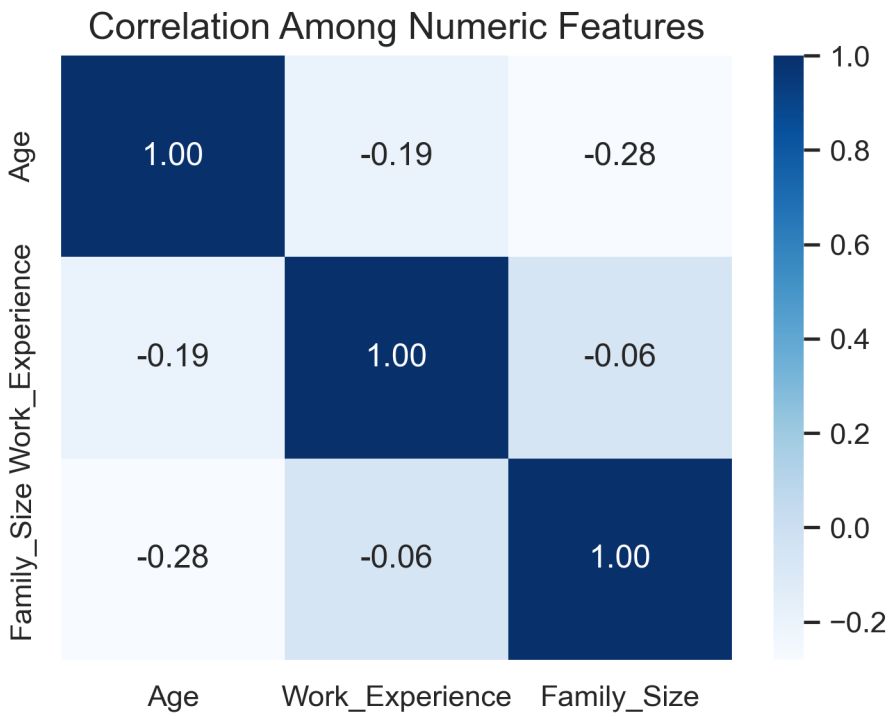


Figure 3: Correlation Among Numeric Features

Training-set class proportions:

Class	Proportion
D	0.281
A	0.244
C	0.244
B	0.23

5. Model Training & Testing

We train multiple supervised classifiers with the same preprocessing pipeline:

- Logistic Regression
- Naive Bayes (Gaussian)
- Decision Tree
- KNN
- Random Forest
- Gradient Boosting
- SVM (linear)
- Neural Network (MLP)

5.1 Test-set Performance (Single Split)

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Gradient Boosting	0.550805	0.534663	0.539623	0.534529
Logistic Regression	0.519827	0.50509	0.509758	0.505425
Neural Network	0.496283	0.47844	0.485774	0.479475
Random Forest	0.490706	0.479511	0.481039	0.47924
KNN	0.482032	0.476488	0.473391	0.473957
Naive Bayes (Gaussian)	0.495663	0.478705	0.482577	0.460877
SVM	0.480173	0.450169	0.467989	0.436654
Decision Tree	0.434944	0.430072	0.427732	0.428699

Best model on the test set (by macro F1): Gradient Boosting

5.2 Stratified 5-Fold Cross-Validation (Train only)

We run CV on the **training set only** to avoid leaking information from the test set during model selection.

Model	CV Accuracy (mean)	CV Accuracy (std)	CV F1 (macro, mean)	CV F1 (macro, std)
Gradient Boosting	0.532539	0.007532	0.517557	0.008018
Logistic Regression	0.504494	0.014851	0.491244	0.015075
Random Forest	0.478462	0.007473	0.467484	0.005069
Neural Network	0.480014	0.009414	0.465288	0.011237
KNN	0.471955	0.007173	0.464183	0.007682
SVM	0.480942	0.015712	0.449134	0.02025
Naive Bayes (Gaussian)	0.471491	0.012329	0.437966	0.016311
Decision Tree	0.435081	0.011443	0.428864	0.011466

Selected best model by CV macro F1: Gradient Boosting

6. Model Selection & Evaluation

6.1 Confusion Matrix (Best Model)

	Pred A	Pred B	Pred C	Pred D
Actual A	178	81	43	92
Actual B	98	128	106	40
Actual C	41	65	234	54
Actual D	79	18	8	349

6.2 Classification Report (Best Model)

```
precision    recall  f1-score   support

           A       0.45      0.45      0.45       394
           B       0.44      0.34      0.39       372
           C       0.60      0.59      0.60       394
           D       0.65      0.77      0.71       454

    accuracy                           0.55      1614
```

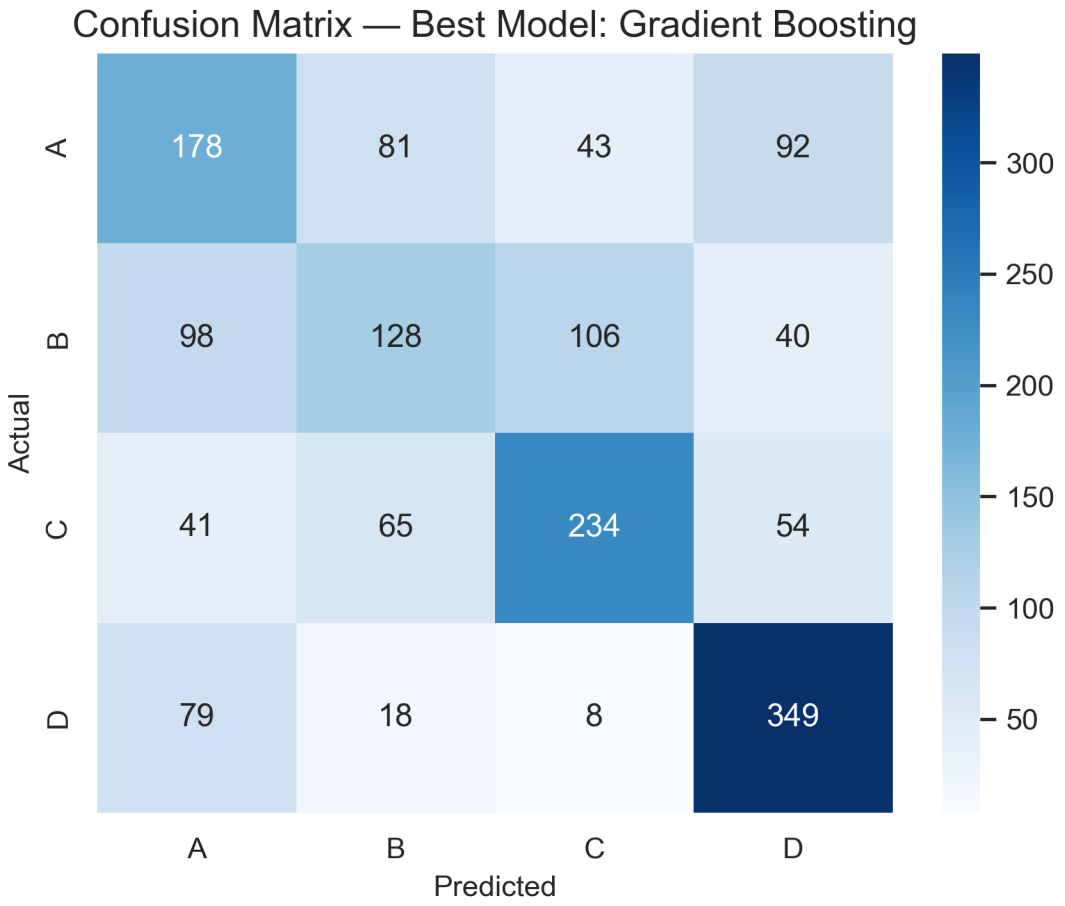


Figure 4: Confusion Matrix — Best Model

```
   macro avg       0.53      0.54      0.53      1614
weighted avg       0.54      0.55      0.54      1614
```

6.3 ROC Curve and AUC (Multi-class OvR)

We compute ROC/AUC using One-vs-Rest (OvR) and report the **macro-average AUC**.

Best model macro-average ROC AUC: 0.795432

AUC comparison across models:

Model	Macro OvR AUC
Gradient Boosting	0.795147
Logistic Regression	0.771516
Neural Network	0.757837
Random Forest	0.750165
Naive Bayes (Gaussian)	0.745834
KNN	0.733704
SVM	0.73249
Decision Tree	0.623049

6.4 Unsupervised Learning: KMeans (Clustering)

KMeans does not use labels; we cluster into 4 groups and compare clusters with the known segmentation labels.

Silhouette score: 0.149725

Cluster vs actual labels (row-normalized):

Actual	0	1	2	3
A	0.125	0.324	0.33	0.221
B	0.095	0.511	0.227	0.167
C	0.112	0.614	0.121	0.153
D	0.465	0.131	0.205	0.199

6.5 Feature Importance (Random Forest)

Feature	Importance
num__Age	0.364494
num__Work_Experience	0.149555
num__Family_Size	0.120375
cat__Profession_Artist	0.03103
cat__Profession_Healthcare	0.029736
cat__Spending_Score_Low	0.025708
cat__Var_1_Cat_6	0.021987
cat__Gender_Male	0.018231
cat__Gender_Female	0.018041
cat__Graduated_No	0.017717
cat__Var_1_Cat_4	0.017332
cat__Ever_Married_Yes	0.016194

Feature	Importance
cat__Graduated_Yes	0.016132
cat__Var_1_Cat_3	0.015669
cat__Ever_Married_No	0.015626

6.6 Example Predictions (New Customers)

ID	Predicted_Segmentation
900001	B
900002	D
900003	C

7. Conclusion

- The dataset has 8068 customers and 4 classes (A/B/C/D).
- Preprocessing handled missing values, encoding, and scaling in a leakage-safe pipeline.
- Based on stratified 5-fold CV (macro F1), the selected best model is **Gradient Boosting**.
- Evaluation used macro metrics, confusion matrix, and ROC/AUC for a full comparison.

This report is generated from the same workflow as the notebook and formatted for printing.

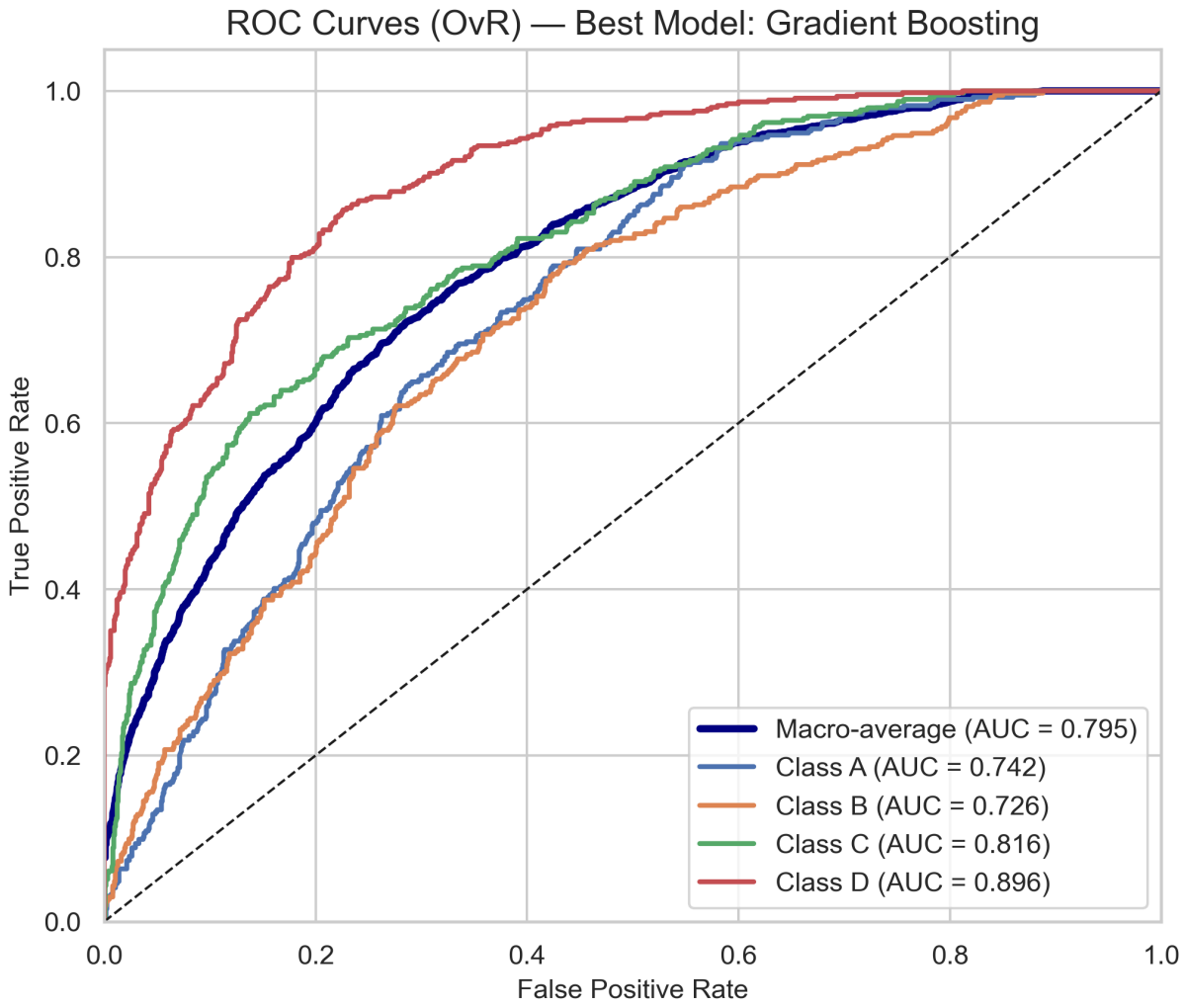


Figure 5: ROC Curves — Best Model

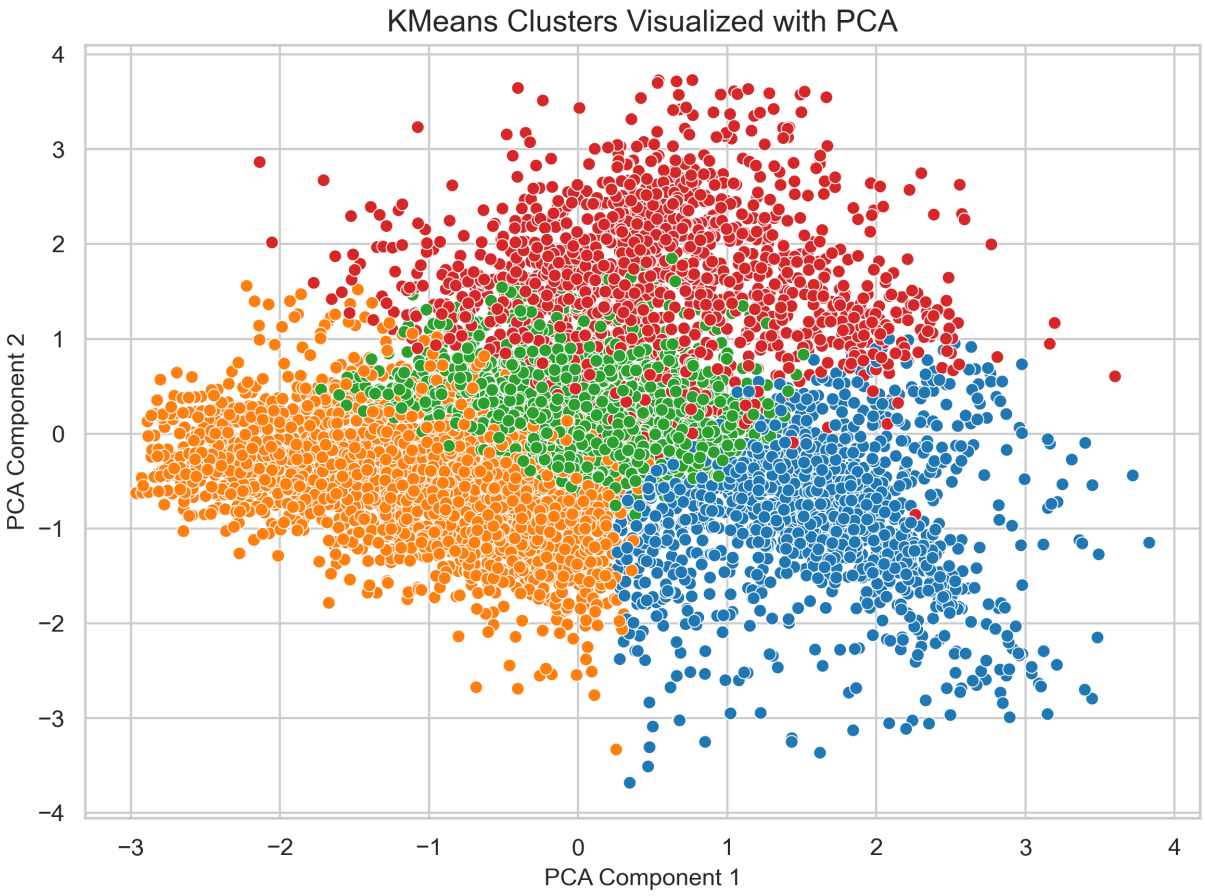


Figure 6: KMeans PCA Visualization

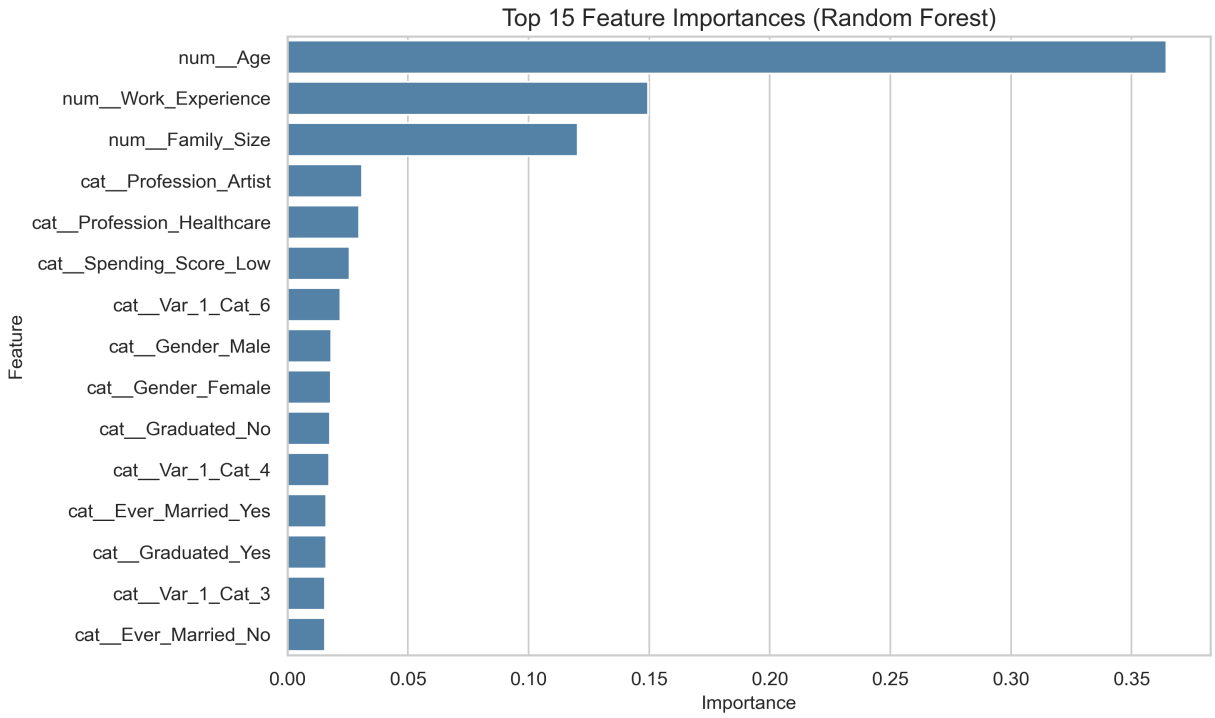


Figure 7: Top Feature Importances